Surface Area of Cuboids

Name _____ **Date** _____ Calculator allowed Version 1

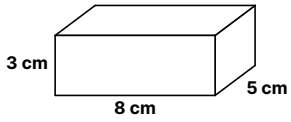
KEY WORDS surface area · face · net · cube · cuboid · open box · square units

A Cubes and cuboids ●○○ — ANSWER ON THE SHEET

Remember: A cuboid has three pairs of equal faces: SA = 2 × (lw + lh + wh). A cube has six equal faces: SA = $6s^2$.

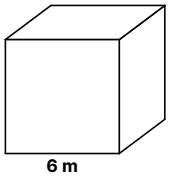
1. Find the surface area of each solid.

a)



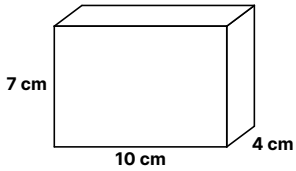
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b)



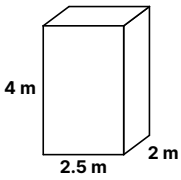
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c)



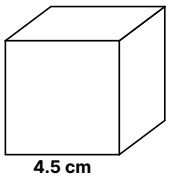
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d)



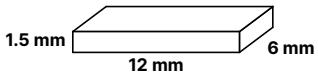
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e)



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f)



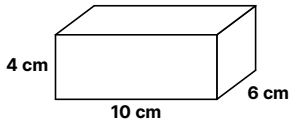
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B Open boxes ●●○ — WORK IN YOUR BOOK

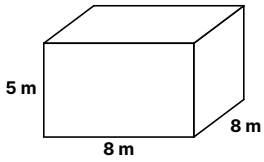
⚠ **Watch out:** An open box has no lid, so count only five faces: the base and the four sides.

2. Each box is open at the top. Find the area of material needed to make it.

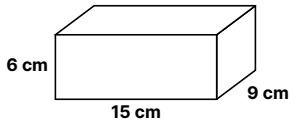
a)



b)



c)



C Working backwards ●●○ — WORK IN YOUR BOOK

Remember: Put the surface area into the formula and solve for the missing length.

3. Find the missing length.

a) A cube has a surface area of 150 cm^2. Find the length of its edges.

b) A cuboid has a base 8 cm by 5 cm and a surface area of 158 cm^2. Find its height.

c) A cube has a surface area of 294 m^2. Find the length of its edges.

d) A cuboid has a base 10 m by 4 m and a surface area of 220 m^2. Find its height.

e) A cube has a surface area of 96 mm^2. Find the length of its edges.

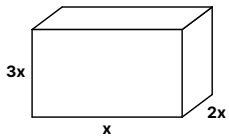
f) A cuboid has a base 5 cm by 3 cm and a surface area of 94 cm^2. Find its height.

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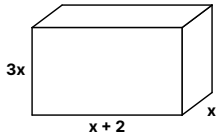
D Lengths with algebra ●●● WORK IN YOUR BOOK

> **Remember:** Write 2 × (lw + lh + wh) with the expressions, expand each product, then collect like terms.

4. A cuboid is x cm by $2x$ cm by $3x$ cm. Write an expression for its surface area and simplify it.



5. A cuboid is $x + 2$ cm by x cm by $3x$ cm. Write an expression for its surface area and simplify it.



E Problems ●●● WORK IN YOUR BOOK

6. A room is 5 m long, 4 m wide and 2.4 m high. The four walls are to be painted, apart from a door and window with a total area of 2.5 m². One litre of paint covers 12 m². How many litres are needed, to 1 decimal place?

7. A gift box is 30 cm by 20 cm by 10 cm. Wrapping paper costs 0.5 cents per cm². Find the cost of paper to cover the box exactly, in dollars.

8. A fish tank with no lid is 1.2 m long, 0.5 m wide and 0.6 m high. Find the area of glass used.

★ Extension FOR EARLY FINISHERS

9. Every edge of a cuboid is doubled. What happens to its surface area, and what happens to its volume?

10. A cuboid with whole-number edges has a volume of 24 cm³. Find the edges that give the smallest surface area, and the ones that give the largest. What do you notice?

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Surface Area of Cuboids

Name _____ **Date** _____ Calculator allowed **Version 2**

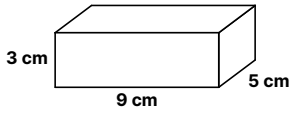
KEY WORDS surface area · face · net · cube · cuboid · open box · square units

A Cubes and cuboids ●○○ — ANSWER ON THE SHEET

Remember: A cuboid has three pairs of equal faces: SA = 2 × (lw + lh + wh). A cube has six equal faces: SA = $6s^2$.

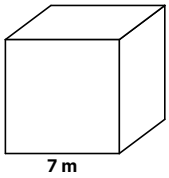
1. Find the surface area of each solid.

a)



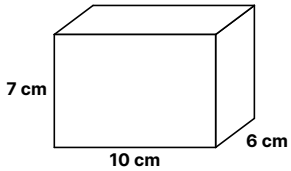
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b)



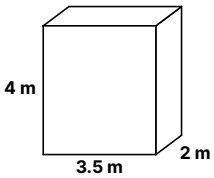
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c)



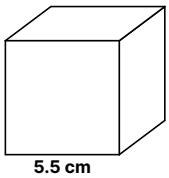
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d)



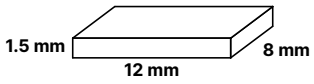
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e)



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f)



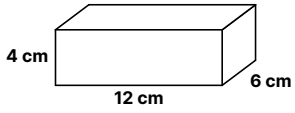
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B Open boxes ●●○ — WORK IN YOUR BOOK

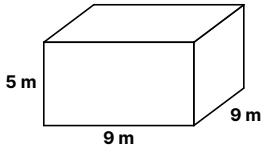
Watch out: An open box has no lid, so count only five faces: the base and the four sides.

2. Each box is open at the top. Find the area of material needed to make it.

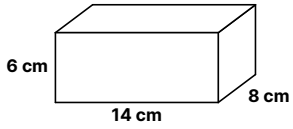
a)



b)



c)



C Working backwards ●●○ — WORK IN YOUR BOOK

Remember: Put the surface area into the formula and solve for the missing length.

3. Find the missing length.

a) A cube has a surface area of 216 cm^2. Find the length of its edges.

b) A cuboid has a base 7 cm by 4 cm and a surface area of 144 cm^2. Find its height.

c) A cube has a surface area of 384 m^2. Find the length of its edges.

d) A cuboid has a base 9 m by 4 m and a surface area of 202 m^2. Find its height.

e) A cube has a surface area of 54 mm^2. Find the length of its edges.

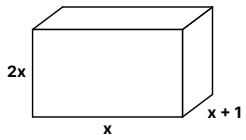
f) A cuboid has a base 6 cm by 2 cm and a surface area of 112 cm^2. Find its height.

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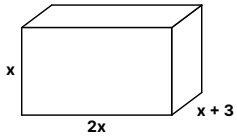
D Lengths with algebra ●●● WORK IN YOUR BOOK

Remember: Write 2 × (lw + lh + wh) with the expressions, expand each product, then collect like terms.

4. A cuboid is x cm by $x + 1$ cm by $2x$ cm. Write an expression for its surface area and simplify it.



5. A cuboid is $2x$ cm by $x + 3$ cm by x cm. Write an expression for its surface area and simplify it.



E Problems ●●● WORK IN YOUR BOOK

6. A room is 6 m long, 4 m wide and 2.5 m high. The four walls are to be painted, apart from a door and window with a total area of 3 m². One litre of paint covers 14 m². How many litres are needed, to 1 decimal place?

7. A gift box is 25 cm by 20 cm by 15 cm. Wrapping paper costs 0.6 cents per cm². Find the cost of paper to cover the box exactly, in dollars.

8. A fish tank with no lid is 1.5 m long, 0.6 m wide and 0.6 m high. Find the area of glass used.

Extension FOR EARLY FINISHERS

9. Every edge of a cuboid is doubled. What happens to its surface area, and what happens to its volume?

10. A cuboid with whole-number edges has a volume of 36 cm³. Find the edges that give the smallest surface area, and the ones that give the largest. What do you notice?

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